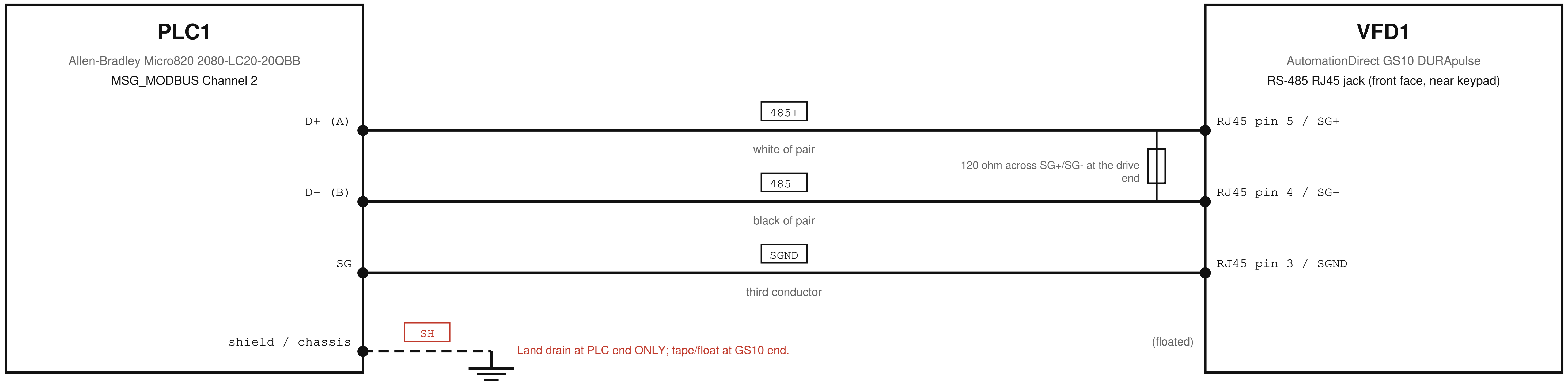
# E-007 RS-485 / MODBUS RTU COMMUNICATION

Conv_Simple / CV-101 (garage conveyor) — Allen-Bradley Micro820 2080-LC20-20QBB (master) ↔ AutomationDirect GS10 DURApulse (node 1) · Modbus RTU only




vfd_comm_ok = TRUE proves the link is up (PLC global).
Read 0x2103 (output freq, Hz x100) via FC03; nonzero readback confirms end-to-end. (Prog_init v2.1:164 'Hz x100'; GS10_UM.txt L15703-05 format XXX.XX Hz)

## CONNECTION TABLE

| Src dev | Src terminal | Dst dev | Dst terminal/pin | Cable / conductor | Wire | Evidence | Notes |
|---|---|---|---|---|---|---|---|
| PLC1 | D+ (A) | VFD1 | RJ45 pin 5 / SG+ | Belden 3105A pair · white of pair | 485+ | verified | Polarity: if silent/garbled, swap 485+/485- first. |
| PLC1 | D− (B) | VFD1 | RJ45 pin 4 / SG− | Belden 3105A pair · black of pair | 485− | verified | |
| PLC1 | SG | VFD1 | RJ45 pin 3 / SGND | 3rd conductor · third conductor | SGND | verified | Dedicated wire, NOT the shield. |
| PLC1 | shield / chassis | VFD1 | (floated) | tinned drain · drain | SH | field_verify | Land drain at PLC end ONLY; tape/float at GS10 end. |

**CCW SERIAL PORT:** Modbus RTU, Master · 9600 · 8N1 (P09.04 = 12) · Channel 2 · Node 1 (P09.00 = 1)

Older GS10_Integration_Guide.md said 8N2 (P09.04=13) - SUPERSEDED by bench-verified 8N1.

Line params pending re-verify (OI-20): 2026-05-20 export read 38.4k/8N2; 2026-05-26 bench sniff read 9600/8N1 (shown). Fresh keypad/readback to adjudicate.

**COMMAND WORDS (0x2000):** STOP = 1 · FWD+RUN = 18 · REV+RUN = 34 · freq register 0x2001

34 NOT 20 — GS10_Integration_Guide.md (2026-03-16) and ControlsToVFD/original PhaseB.st print 20; corrected by PhaseB_V1.4 'FIX 2' (bit 5 REV + bit 1 Run), Beginner_Verify_V2 ('NOT 20!'), WI-001 Table 9, and live Prog_init v2.1 (vfd_cmd_word := 34).

Beginner_Verify_V2 §4.4 (GS10 manual p.6); WI-001 Tables 8-10; PhaseB_V1.4.st; Prog_init_ConvSimple_v2.1.st

CORRECTED from May-16 draft: Channel 0 → 2 · SGND pin 1/8 → pin 3 · 8N2 → 8N1. Do not copy the old values.

## NOTES

- Belden 3105A — shielded twisted pair (485+/485-) + 1 conductor (SGND) + drain
- VERIFIED (solid) = recovered CommsToVFD + Beginner_Verify p48 + Prog_init v2.1; FIELD VERIFY (dashed) = exact PLC chassis ground point; cable P/N if not Belden 3105A.
- Port locations per CommsToVFD (PLC top-edge connector; GS10 front-face RJ45) — WI-001 says bottom-front. Wayfinding only; confirm on unit (FIELD VERIFY).
- GS10 RS-485 RJ45 cable confirmed landed (photo wire_4). The GS10 control connector (FWD/REV/DI/AI/AO1/DO1) is also wired — see OI-22.
- GS10 RS-485 RJ45 confirmed landed (full-res photo). The GS10 top run/DI control connector is EMPTY — no hardwired run/dir; command is Modbus. Bottom analog/output row is wired (OI-22).
- GS10 run/freq source = Modbus: P00.20=1, P00.21=2 (2026-05-20 export; same fact as E-006).
- P09.09 (Response Delay) = 10.0 ms — program flags CRITICAL (default 2.0 ms → ErrorID-55): Prog_init_ConvSimple_v2.1.st:79.
- 120 ohm termination (SG+/SG- at GS10 end): often omitted on a <2 m bench run — presence is FIELD VERIFY, not confirmed here.

**SOURCES:**
full-res photos review/photos/*_full.jpg + crops (2026-07-11); review/PHOTO_EVIDENCE_V5.md

**DOC REFS:**
CCW/MIRA_PLC/docs/instructions/Conv_Simple_CommsToVFD.pdf §2 (recovered style + wiring)
CCW/MIRA_PLC/docs/instructions/Conv_Simple_GS10_Beginner_Verify_V2.pdf p48 (corrected pins/channel)
plc/Prog_init_ConvSimple_v2.1.st (Channel 2, node 1, 0x2103)
plc/GS10_Integration_Guide.md §3 (RJ45 pinout, register map)

## TROUBLESHOOTING (this circuit)

- Nothing after config -> swap 485+/485- (A/B polarity; vendors disagree).
- CRC errors -> usually polarity. Complete silence -> usually baud mismatch (confirm 9600 8N1).
- GS10 CE10 / F30 comm-timeout -> no Modbus packets; check wiring, baud, P09.03 timeout.
- 120 ohm termination across SG+/SG- at the GS10 end for long runs (bench <2 m usually fine).
- SGND is a dedicated conductor, not the shield; missing common -> intermittent corruption.

## LEGEND

VERIFIED (solid)
FIELD VERIFY (dashed)
Wxxx proposed wire number (red = unverified)

| | |
|---|---|
| RS-485 / MODBUS RTU<br>Conv_Simple / CV-101 (garage conveyor) | SHEET<br>E-007 |
| Drawn: MIRA / FactoryLM<br>recovers MIRA-WI-001 / Conv_Simple_CommsToVFD §2 | REV A 7 of 9<br>2026-07-11 |